LESSON 01 / 03

ASSUMPTIONS & CONVENTIONS

STATUS — olympiad/specialised extension (IOQM/RMO-flavoured)

Lattice-path methods, Catalan numbers and reflection arguments are beyond the official P&C syllabus lines; they are the standard Olympiad continuation of $\binom{n}{r}$ as a path count.

KEY RESULT

Paths are binomial coefficients

Monotone paths from $(0,0)$ to (a,b) (steps right/up): choose which steps are up: $\binom{a+b}{b}$. Paths that never rise above the diagonal $y=x$ to (n,n): the **Catalan number**

$$C_n = \binom{2n}{n} - \binom{2n}{n+1} = \frac{1}{n+1}\binom{2n}{n} = 1, 1, 2, 5, 14, 42, \ldots$$

Reflection principle: a "bad" path first touching $y=x+1$, reflected up to that point, bijects with all paths from $(-1,1)$ to (n,n) — $\binom{2n}{n-1}$ of them. Subtract. Paths to $(4,4)$ staying weakly below the diagonal: $C_4 = 70 - 56 = 14$.

KEY RESULT

Ballot theorem

Votes p for A, q for B, $p>q$: counting orders where A is **strictly ahead throughout**: $\frac{p-q}{p+q}\binom{p+q}{q}$. For $p=5$, $q=3$: $\frac{2}{8}\cdot 56 = 14$. The same reflection gives paths to (a,b), $a \geq b$, never above the diagonal: $\frac{a-b+1}{a+1}\binom{a+b}{b}$.

INTUITION

Why Catalan counts everything

Dyck paths, balanced parentheses, full binary trees, polygon triangulations, non-crossing matchings share one decomposition: *first return splits the object into inside × outside*, giving $C_{n+1} = \sum_{i=0}^{n} C_i C_{n-i}$. Recognising the convolution shape is the solution; the closed form follows. Brute-force checks: Dyck paths of semilength $n \leq 7$ counted by enumeration in math_checks.

COMMON TRAP

Common traps — path counts

- "Never above" vs "never below" and weak vs strict inequalities change which line to reflect in ($y=x+1$ vs $y=x-1$): draw the first forbidden point.
- Ballot needs $p>q$ strictly; ties throughout (weakly ahead) is the Catalan-style variant with a different formula.

LESSON 02 / 03

ASSUMPTIONS & CONVENTIONS

STATUS — olympiad/specialised extension (IOQM/RMO-flavoured)

Group-action counting is university/Olympiad material. It exists in this chapter because Part I's "divide by the symmetry" is exactly the special case where every orbit has the same size — and fails precisely when it does not.

SHORT DERIVATION

Burnside's lemma (double counting in disguise)

A group G acts on colourings X; orbits = essentially-different colourings. Count pairs (g, x) with $gx = x$ two ways: by g it is $\sum_g |\mathrm{Fix}(g)|$; by x it is $\sum_x |\mathrm{Stab}(x)| = \sum_x \frac{|G|}{|\mathrm{orbit}(x)|}$, which sums to $|G| \times$ (number of orbits). Hence

$$\#\text{orbits} = \frac{1}{|G|} \sum_{g \in G} |\mathrm{Fix}(g)|$$

A colouring is fixed by g iff it is constant on g's cycles: $k^{\#\text{cycles}}$ colourings in k colours.

WORKED EXAMPLE

Worked — necklaces and bracelets

Necklaces of n beads, k colours (rotations only): $N(n,k) = \frac{1}{n} \sum_{d \mid n} \varphi(d)\, k^{n/d}$ (rotations grouped by $\gcd(n, s)$). $N(4,2) = \frac{1}{4}(16 + 4 + 4) = 6$: RRRR, RRRB, RRBB, RBRB, RBBB, BBBB ✓. Bracelets of 6 beads, 2 colours (full dihedral group of order 12): fixed counts $64 + 2 \cdot 2 + 2 \cdot 4 + 8 + 3 \cdot 16 + 3 \cdot 8 = 156$, orbits $156/12 = 13$ ✓.

COMMON TRAP

Common traps — Burnside

- Forgetting the identity element (usually the largest Fix count).
- Grouping rotations wrongly: rotation by s on n beads has $\gcd(n, s)$ cycles, and there are $\varphi(n/d)$ rotations with a given divisor structure — recount small cases by listing (as with $N(4,2) = 6$).
- Applying "divide by $|G|$" directly to colourings: that is exactly the non-exact symmetry error Burnside repairs.

LESSON 03 / 03

ASSUMPTIONS & CONVENTIONS

STATUS — olympiad/specialised extension (IOQM/RMO-flavoured)
Integer partitions, Stirling/Bell numbers and extremal combinatorics (Sperner, Erdős–Ko–Rado) are Olympiad territory beyond the JEE P&C lines.

KEY RESULT

Integer partitions: no closed form, one generating function
$p(n)$ counts partitions of n (order ignored): $1, 2, 3, 5, 7, 11, 15, 22, 30, 42$ for $n = 1..10$. There is no stars-and-bars model; the count lives in

$$\sum_{n \geq 0} p(n)x^n = \prod_{k \geq 1} \frac{1}{1-x^k}$$

Euler: partitions into **distinct** parts = partitions into **odd** parts, since $\prod(1+x^k) = \prod \frac{1-x^{2k}}{1-x^k} = \prod_{m \text{ odd}} \frac{1}{1-x^m}$ (for $n = 10$: 10 of each, listed in math_checks by enumeration). Conjugation of Young diagrams: partitions into at most k parts ↔ partitions with largest part $\leq k$. Euler's pentagonal theorem gives $p(n) = p(n-1) + p(n-2) - p(n-5) - p(n-7) + \cdots$ (check $p(10) = 30 + 22 - 7 - 3 = 42$ ✓).

KEY RESULT

Stirling numbers of the second kind and Bell numbers
$S(n,k)$ = partitions of an n-set into k nonempty **unlabelled** blocks: recurrence ("where does element n go?") $S(n,k) = kS(n-1,k) + S(n-1,k-1)$; closed form by IE $S(n,k) = \frac{1}{k!}\sum_j (-1)^j \binom{k}{j}(k-j)^n$, because $k!\,S(n,k)$ counts onto functions.

$S(6,3) = 3 \cdot 25 + 15 = 90 = \frac{729 - 192 + 3}{6}$ ✓. Bell numbers $B_n = \sum_k S(n,k)$: $1, 1, 2, 5, 15, 52, 203$. Dividing 6 people into 3 unlabelled pairs: $\frac{6!}{(2!)^3 3!} = 15$.

KEY RESULT

Extremal families: Sperner and Erdős–Ko–Rado
Sperner: an antichain of subsets of an n-set (no one containing another) has at most $\binom{n}{\lfloor n/2 \rfloor}$ members — the middle layer is extremal (LYM proof: a random maximal chain meets an antichain at most once, and hits a k-set with probability $1/\binom{n}{k}$).

Erdős–Ko–Rado: for $n \geq 2k$, an intersecting family of k-subsets of an n-set has at most $\binom{n-1}{k-1}$ members; the star (all sets through one fixed point) attains it. Hypothesis essential: for $n < 2k$ every two k-sets meet and the whole $\binom{n}{k}$ is intersecting.

COMMON TRAP

Common traps — advanced counting

- Confusing partitions (no formula) with compositions ($\binom{n-1}{k-1}$): "order ignored" changes the universe, not just the count.
- $S(n,k)$ vs onto functions: the factor $k!$ is the labelling of blocks.
- Quoting EKR/Sperner without their hypotheses ($n \geq 2k$; antichain, not intersecting).